VANISHING OF MAPS BETWEEN CARTESIAN DERIVED OBJECTS UNDER NEGATIVE-EXT CONDITIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the objects-twin of lemma-nullity-cartesian-modules-derived (cited in card L46179's uniqueness argument); the identification X_n = g_{n!}K_n is established on card L46169 (lemma-derived-cartesian-shriek); no verification is omitted.

Editorial note. The label lemma-nullity-cartesian-objects-derived names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The Postnikov system associated to a derived cartesian object. Same chapter, \label{lemma-abelian-postnikov}. Proof lines 3062-3064: the objects X_n and the Postnikov system Y_n associated to K, with K = hocolim Y_n[n].
- Cartesian derived objects are g_{n!} of their components. Same chapter, \label{lemma-derived-cartesian-shriek} (card L46169). Proof lines 3064-3066: X_n = g_{n!}K_n by cartesianity of K, so Y_0 = X_0 = g_{0!}K_0.
- The homotopy colimit triangle. Same chapter, \label{lemma-abelian-postnikov} (defining triangle of the hocolim). Proof lines 3083-3090: the distinguished triangle with the direct sums defining the homotopy colimit, used for the final reduction.

Source reference. Original reference 0D9I: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3002–3002. Prerequisite pointer invoked at proof line(s) 3060; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9E: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2752–2752. Prerequisite pointer invoked at proof line(s) 3063; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 3047; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE. If $K, K' \in D(\mathcal{C}_{total})$. Assume*

(1) *K is cartesian,*
(2) *$\mathrm{Hom}(K_i[i], K'_i) = 0$ for $i > 0$, and*
(3) *$\mathrm{Hom}(K_i[i+1], K'_i) = 0$ for $i \geq 0$.*

Then any map $K \to K'$ which induces the zero map $K_0 \to K'_0$ is zero.

Proof. Consider the objects X_n and the Postnikov system Y_n associated to K in Lemma 0D9I. As $K = \mathrm{hocolim} Y_n[n]$ the map $K \to K'$ induces a compatible family of morphisms $Y_n[n] \to K'$. By (1) and Lemma 0D9E we have $X_n = g_{n!}K_n$. Since $Y_0 = X_0$ we find that $K_0 \to K'_0$ being zero implies $Y_0 \to K'$ is zero. Suppose we've shown that the map $Y_n[n] \to K'$ is zero for some $n \geq 0$. From the distinguished triangle

$$Y_n[n] \to Y_{n+1}[n+1] \to X_{n+1}[n+1] \to Y_n[n+1]$$

we get an exact sequence

$$\mathrm{Hom}(X_{n+1}[n+1], K') \to \mathrm{Hom}(Y_{n+1}[n+1], K') \to \mathrm{Hom}(Y_n[n], K')$$

As $X_{n+1}[n+1] = g_{n+1!}K_{n+1}[n+1]$ the first group is equal to

$$\mathrm{Hom}(K_{n+1}[n+1], K'_{n+1})$$

which is zero by assumption (2). By induction we conclude all the maps $Y_n[n] \to K'$ are zero. Consider the defining distinguished triangle

$$\bigoplus Y_n[n] \to \bigoplus Y_n[n] \to K \to (\bigoplus Y_n[n])[1]$$

for the homotopy colimit. Arguing as above, we find that it suffices to show that

$$\mathrm{Hom}((\bigoplus Y_n[n])[1], K') = \prod \mathrm{Hom}(Y_n[n+1], K')$$

is zero for all $n \geq 0$. To see this, arguing as above, it suffices to show that

$$\mathrm{Hom}(K_n[n+1], K'_n) = 0$$

for all $n \geq 0$ which follows from condition (3). □